

SQL Exercise - Stored procedure

Employee Management System SQL Exercises

```
CREATE DATABASE IF NOT EXISTS EmployeeManagement;

USE EmployeeManagement;

;

CREATE TABLE Departments (
    DepartmentID INT PRIMARY KEY,
    DepartmentName VARCHAR(100)
);

CREATE TABLE Employees (
    EmployeeID INT PRIMARY KEY,
    FirstName VARCHAR(50),
    LastName VARCHAR(50),
    DepartmentID INT,
    Salary DECIMAL(10,2),
    JoinDate DATE,
    FOREIGN KEY (DepartmentID)
        REFERENCES Departments(DepartmentID)
);

INSERT INTO Departments (DepartmentID, DepartmentName)
VALUES
(1, 'HR'),
(2, 'Finance'),
(3, 'IT'),
(4, 'Marketing');

INSERT INTO Employees
(EmployeeID, FirstName, LastName, DepartmentID, Salary, JoinDate)
VALUES
(1, 'John', 'Doe', 1, 5000.00, '2020-01-15'),
```

```
(2, 'Jane', 'Smith', 2, 6000.00, '2019-03-22'),
(3, 'Michael', 'Johnson', 3, 7000.00, '2018-07-30'),
(4, 'Emily', 'Davis', 4, 5500.00, '2021-11-05');
```

```
SELECT * FROM Departments;
```

```
SELECT * FROM Employees;
```

Result Grid		Filter Rows:		Edit:		Export/Import:	
	EmployeeID	FirstName	LastName	DepartmentID	Salary	JoinDate	
▶	1	John	Doe	1	5000.00	2020-01-15	
	2	Jane	Smith	2	6000.00	2019-03-22	
	3	Michael	Johnson	3	7000.00	2018-07-30	
	4	Emily	Davis	4	5500.00	2021-11-05	
	5	Robert	Brown	3	8000.00	2024-01-10	
✱	NULL	NULL	NULL	NULL	NULL	NULL	

```
-- =====
```

-- EXERCISE 1

-- CREATE PROCEDURE TO RETRIEVE EMPLOYEES BY DEPARTMENT

```
-- =====
```

```
DELIMITER $$
```

```
CREATE PROCEDURE GetEmployeesByDepartment(
```

```
    IN p_DepartmentID INT
```

```
)
```

```
BEGIN
```

```
    SELECT EmployeeID,
```

```
           FirstName,
```

```
           LastName,
```

```
           DepartmentID,
```

```
           Salary,
```

```
           JoinDate
```

```
    FROM Employees
```

```
    WHERE DepartmentID = p_DepartmentID;
```

```
END $$
```

```
DELIMITER ;
```

```
CALL GetEmployeesByDepartment(3);
```

Result Grid						
Filter Rows:		Export:		Wrap Cell Content:		
	EmployeeID	FirstName	LastName	DepartmentID	Salary	JoinDate
▶	3	Michael	Johnson	3	7000.00	2018-07-30
	5	Robert	Brown	3	8000.00	2024-01-10

DELIMITER \$\$

CREATE PROCEDURE sp_InsertEmployee(

IN p_EmployeeID INT,

IN p_FirstName VARCHAR(50),

IN p_LastName VARCHAR(50),

IN p_DepartmentID INT,

IN p_Salary DECIMAL(10,2),

IN p_JoinDate DATE

)

BEGIN

INSERT INTO Employees

(

EmployeeID,

FirstName,

LastName,

DepartmentID,

Salary,

JoinDate

)

VALUES

(

p_EmployeeID,

p_FirstName,

p_LastName,

p_DepartmentID,

p_Salary,

p_JoinDate

```

);
END $$
DELIMITER ;
CALL sp_InsertEmployee(
    5,
    'Robert',
    'Brown',
    3,
    8000.00,
    '2024-01-10'
);
-- Check inserted employee
SELECT * FROM Employees;

```

Result Grid						
Filter Rows:						
Edit:    Export/Import: 						
	EmployeeID	FirstName	LastName	DepartmentID	Salary	JoinDate
▶	1	John	Doe	1	5000.00	2020-01-15
	2	Jane	Smith	2	6000.00	2019-03-22
	3	Michael	Johnson	3	7000.00	2018-07-30
	4	Emily	Davis	4	5500.00	2021-11-05
	5	Robert	Brown	3	8000.00	2024-01-10
✱	NULL	NULL	NULL	NULL	NULL	NULL

```
-- =====
```

-- EXERCISE 4

-- EXECUTE STORED PROCEDURE FOR DEPARTMENT 2

```
-- =====
```

```
CALL GetEmployeesByDepartment(2);
```

Result Grid						
Filter Rows:						
Export:  Wrap Cell Content: 						
	EmployeeID	FirstName	LastName	DepartmentID	Salary	JoinDate
▶	2	Jane	Smith	2	6000.00	2019-03-22

```
-- =====
```

-- EXERCISE 5

-- STORED PROCEDURE TO RETURN EMPLOYEE COUNT

-- =====

DELIMITER \$\$

CREATE PROCEDURE GetEmployeeCount(

IN p_DepartmentID INT

)

BEGIN

SELECT

DepartmentID,

COUNT(*) AS TotalEmployees

FROM Employees

WHERE DepartmentID = p_DepartmentID

GROUP BY DepartmentID;

END \$\$

DELIMITER ;

CALL GetEmployeeCount(3);

Result Grid		Filter Rows:	Ex
	DepartmentID	TotalEmployees	
▶	3	2	

SHOW PROCEDURE STATUS

WHERE Db = 'EmployeeManagement';

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Db	Name	Type	Definer	Modified	Created
▶	employeemanagement	GetEmployeeCount	PROCEDURE	root@localhost	2026-06-16 09:02:54	2026-06-16 09:02:54
	employeemanagement	GetEmployeesByDepartment	PROCEDURE	root@localhost	2026-06-16 09:02:54	2026-06-16 09:02:54
	employeemanagement	sp_InsertEmployee	PROCEDURE	root@localhost	2026-06-16 09:02:54	2026-06-16 09:02:54